ALPHA SYSTEM REPLACEMENT

GDPS Hardware Infrastructure, Cluster, Storage, and Replication Master Design

DEV • TEST • PROD • DR • COOP

Hardware Baseline for the Alpha/OpenVMS Replacement

Table 1 identifies the Alpha System Replacement project, the GDPS system name, document title, security classification, and current engineering/provisioning status. It establishes the administrative identity of the hardware master before the technical content begins.

Table 1

Document Control

Project	**PR-26-GPO-2540 — Alpha System Replacement**
System	GDPS
Document	Hardware Infrastructure Master Design
Classification	UNCLASSIFIED
Status	Infrastructure Baseline v3.7 - Cross-Data-Center Failover

This document establishes the replacement hardware, cluster, storage, replication, SMB namespace, and recovery infrastructure required to transition the existing Alpha environment to GDPS.

Executive Summary

GDPS replaces the existing Alpha/OpenVMS infrastructure while preserving the operational characteristics on which current congressional publishing workflows and legacy Windows applications depend. The replacement is intentionally structured as a lift-and-shift of the existing availability, shared-storage, replication, and file-service behavior rather than a redesign of the underlying business process.

The target platform is divided into five environments: DEV, TEST, PROD, DR, and COOP. DEV is implemented as a four-VM development stack on the existing GPO virtualization platform and continues to use external enterprise NAS storage. TEST is the new pre-production environment that Alpha does not have today and is intentionally designed to mirror PROD so clustering, failover, shared storage, replication, application deployment, and recovery behavior can be exercised before a production release.

PROD replaces the current ALPHA3 Production environment. DR separately replaces and improves the ALPHA6 Production Backup / disaster-recovery capability. TEST, PROD, DR, and COOP each preserve the proven two-node physical file/service cluster pattern on supported Windows Server Failover Clustering. HQ and IM are separate GPO data centers. TEST and PROD deliberately place one physical cluster node at HQ and the peer node at IM so the replacement can tolerate not only a server-node failure but also the loss of either data center, subject to quorum, storage, and inter-site network availability. COOP preserves the self-contained recovery model represented by the existing Data Center in a Crate and includes its own two-node physical file/service cluster, enterprise crate storage, and replicated production data.

Legacy SMB compatibility is a mandatory infrastructure requirement. Existing applications currently consume data through UNC interfaces such as \\alpha0\<share>, \ \alpha3\<share>, and \\alpha6\<share>. Those application interfaces cannot be changed during this migration. GDPS therefore preserves alpha0, alpha3, and alpha6 as clustered SMB client-access identities after cutover, while the physical GDPS cluster nodes retain their permanent GDPS server names.

Two-node TEST, PROD, DR, and COOP physical file/service clusters use a File Share Witness for quorum. The witness is independent of the clustered business-data storage. Temporary SMB identities support staging and validation before cutover; legacy-compatible production identities are activated only after the corresponding legacy Alpha identities have been disabled.

Key Infrastructure Decisions

Table 2 summarizes the principal infrastructure decisions that govern the GDPS replacement. It highlights the requirement for TEST to mirror PROD, the two-node clustered file-service pattern, File Share Witness quorum, production and COOP storage capacities, and preservation of both legacy and temporary SMB identities.

Table 2

Key Infrastructure Decisions

Area	GDPS Baseline	Purpose
TEST	Mirrors PROD	Validate production topology before deployment
File service	Two-node physical WSFC in TEST, PROD, DR, COOP	Preserve the confirmed two-node Alpha cluster availability baseline
Quorum	File Share Witness	Provide third vote for two-node clusters
PROD shared storage	14.4 TB usable minimum	Preserve current Alpha production storage baseline
COOP shared storage	21.4 TB usable minimum	Preserve current crate storage baseline
Legacy SMB names	alpha0 / alpha3 / alpha6	Preserve unchanged legacy UNC consumers
Migration SMB names	alpha0-1 / alpha3-1 / alpha6-1	Allow staging without production identity conflict
DEV hosting	Four VMs on existing GPO virtualization infrastructure	Avoid unnecessary physical DEV acquisition while retaining complete development roles
DR	Two-node physical recovery file/service cluster	Replace and improve ALPHA6 Production Backup / DR capability

Table of Contents

Table 3 lists the document sections and their page locations so reviewers can navigate directly to the legacy baseline, replacement architecture, server matrices, storage, replication, cluster standards, and SMB namespace controls.

Table 3

Table of Contents

Section	Page
Executive Summary	1
1. Purpose	3
2. Existing Alpha Hardware Baseline	4
2.1 Confirmed OpenVMS Cluster Composition	4
2.2 Legacy Alpha Environment Diagram	6
2.3 Legacy Production Cluster Components	7
2.4 Legacy COOP Data Center in a Crate	8
2.5 Legacy Alpha3 Disk Volume Allocation	9
3. GDPS Replacement Hardware Architecture	10
4. Hardware Server Matrices	11
4.1 DEV Environment	11
4.2 TEST Environment (Production Mirror)	12
4.3 PROD, DR, and COOP Environments	14
5. Disk Volume / GDPS Volume Mapping	17
6. Shared Storage Baseline	18
7. Replication and Failover Architecture	18
8. Legacy-to-GDPS Hardware Mapping	19
9. Provisioning Summary	20
10. Infrastructure Guardrails	21
11. GDPS Windows File Cluster Standard	22
11.1 File Share Witness Baseline	22
12. Legacy SMB Namespace Preservation	23
12.1 SMB Namespace Cutover Sequence	23
12.2 SMB Usage Boundary	24
13. Additional Infrastructure Guardrails	25

1. Purpose

Documentation rule: every substantive table and diagram in this hardware master is assigned a sequential APA-style Table or Figure number and descriptive title. Each is also accompanied by explanatory prose describing what is depicted, why it matters, and how it relates to the Alpha-to-GDPS lift-and-shift baseline. These identifiers shall be used for subsequent discussion and cross-reference.

This document establishes the hardware and infrastructure baseline for the GDPS replacement of the existing Alpha/OpenVMS environment. The design preserves the existing clustered file/service availability, shared enterprise storage, data replication, production failover, and COOP recovery capabilities while replacing unsupported Alpha/OpenVMS technology with the new GDPS platform.

The replacement is a lift-and-shift of existing operational behavior. POLLVAX remains the authority for the business behavior being ported. The infrastructure design therefore preserves current availability and replication semantics rather than redesigning them.

2. Existing Alpha Hardware Baseline

Table 4 depicts the known current-state Alpha/OpenVMS hardware baseline by system. It identifies the operational role, Integrity server generation, shared-storage platform, operating status, and the availability or replication purpose associated with ALPHA0, ALPHA3, ALPHA6, ALPHA10, and ALPHA13.

Table 4

Existing Alpha Hardware Baseline

Legacy System	Role	Compute	Shared Storage	Status	Availability / Replication Purpose
Alpha0 / DCIAC1	Development	rx2800 i2	3PAR 7200	Out of support	Receives replicated production data where currently configured
Alpha3	Production	rx2800 i6	3PAR 8200	Production	Primary production cluster / authoritative production data
Alpha6	Production Backup System / DR	rx2800 i6	3PAR 8200	Production backup / DR	Production backup and disaster-recovery capability
Alpha10 / DCIAC2	DCIAC02	rx2800 i4	3PAR 8200	COOP / DCIAC	Data Center In-a-Crate recovery capability
Alpha13 / DCIAC3	DCIAC03 / Data Center In-a-Crate	rx2800 i6	HPE Primera	COOP / DCIAC	Data Center In-a-Crate; receives production data

2.1 Confirmed OpenVMS Cluster Composition

The following current-state cluster composition was confirmed directly by the Alpha administrator. These node relationships are authoritative for the physical-cluster baseline used by the GDPS replacement design.

Table 5 records the Alpha administrator-confirmed composition of the major OpenVMS clusters. It shows that ALPHA3, ALPHA6, ALPHA10, and ALPHA13 each consist of two physical nodes and identifies their confirmed production, backup/DR, or DCIAC roles and locations.

Table 5

Confirmed OpenVMS Cluster Composition

Current Alpha Cluster	Confirmed Role	Physical Nodes	Location
ALPHA3	Production	N3A + N3B = 2 nodes	HQ
ALPHA6	Production Backup System	N6A + N6B = 2 nodes	IMDC
ALPHA10	DCIAC02	ALPHA11 + ALPHA12 = 2 nodes	COOP / DCIAC
ALPHA13	DCIAC03 / Data Center In-a-Crate	N13A + N13B = 2 nodes	COOP / DCIAC

ALPHA0	Development	ALPHA1 + ALPHA2 = 2 nodes	Legacy DEV / DCIAC1

Legacy Two-Node Cluster Baseline for GDPS

The confirmed Alpha administrator inventory establishes a consistent two-node physical cluster pattern for the current Production, Production Backup/DR, and DCIAC environments. ALPHA3, ALPHA6, ALPHA10, and ALPHA13 each consist of two physical OpenVMS nodes. This is the current-state availability baseline for the GDPS replacement file/service tier.

Table 6 translates the confirmed legacy two-node cluster pattern into the GDPS replacement baseline. It shows that TEST, PROD, DR, and COOP retain two-node physical file/service clusters, while DEV is intentionally implemented as a four-VM development stack on existing GPO virtualization infrastructure. This DEV efficiency improvement does not alter the production, recovery, or COOP physical-cluster standard.

Table 6

Legacy Two-Node Cluster Baseline for GDPS

Current Alpha Cluster	Current Physical Baseline	Confirmed Role	GDPS File/Service Baseline	Lift-and-Shift / Improvement Rationale
ALPHA3	N3A + N3B = 2 nodes	Production	Two-node GDPS PROD file/service cluster	Directly preserves the current two-node Production cluster baseline on supported infrastructure.
ALPHA6	N6A + N6B = 2 nodes	Production Backup System / DR	Two-node recovery/file-service capability	Preserves the current two-node Production Backup/DR baseline; recovery may be improved without reducing capability.
ALPHA10	ALPHA11 + ALPHA12 = 2 nodes	DCIAC02	Two-node COOP/recovery file/service capability	Preserves the current two-node DCIAC recovery pattern.
ALPHA13	N13A + N13B = 2 nodes	DCIAC03 / Data Center In-a-Crate	Two-node COOP crate file/service cluster	Directly preserves the two-node crate cluster baseline while modernizing storage, failover, and supportability.
No dedicated Alpha TEST	None	No dedicated TEST environment today	Two-node GDPS TEST file/service cluster mirroring PROD	Intentional improvement: TEST mirrors PROD so the proven two-node production topology can be rehearsed before production deployment.
ALPHA0	ALPHA1 + ALPHA2 = 2 nodes	Development	GDPS DEV uses four VMs on GPO virtualization infrastructure	Preserves DEV functional roles while intentionally reducing physical hardware; this does not weaken the physical cluster baseline for TEST/PROD/DR/COOP.

BASELINE RULE: GDPS shall not reduce the confirmed two-node clustered availability pattern for TEST, PROD, DR, or COOP. The two-node Windows Server Failover Cluster used for these GDPS file/service tiers is a direct replacement baseline, not newly invented redundancy. TEST intentionally mirrors the PROD two-node pattern so the production design can be exercised before deployment. DEV is the explicit exception and is virtualized because it is a development environment rather than the production availability benchmark.

CROSS-DATA-CENTER AVAILABILITY RULE: HQ and IM are independent data centers. For TEST and PROD, the two physical file/service nodes shall be split across HQ and IM. The design must be validated for (1) loss of a single node, (2) loss of the HQ data center, and (3) loss of the IM data center. Successful site failover depends on the surviving site's network, storage/data availability, quorum, and witness reachability.

ALPHA0/DEV is confirmed as a two-node legacy cluster (ALPHA1 and ALPHA2). GDPS DEV does not reproduce that physical node count because DEV is intentionally hosted as virtual machines on the existing GPO virtualization platform; the production-grade two-node physical baseline is preserved in TEST, PROD, DR, and COOP.

Source: Alpha administrator confirmation supplied by the project lead; corroborated by the 08AUG2025 ALPHA System Storage Information deck for system roles.

2.2 Legacy Alpha Environment Diagram

Figure 1

Legacy Alpha Environment and Replication Topology

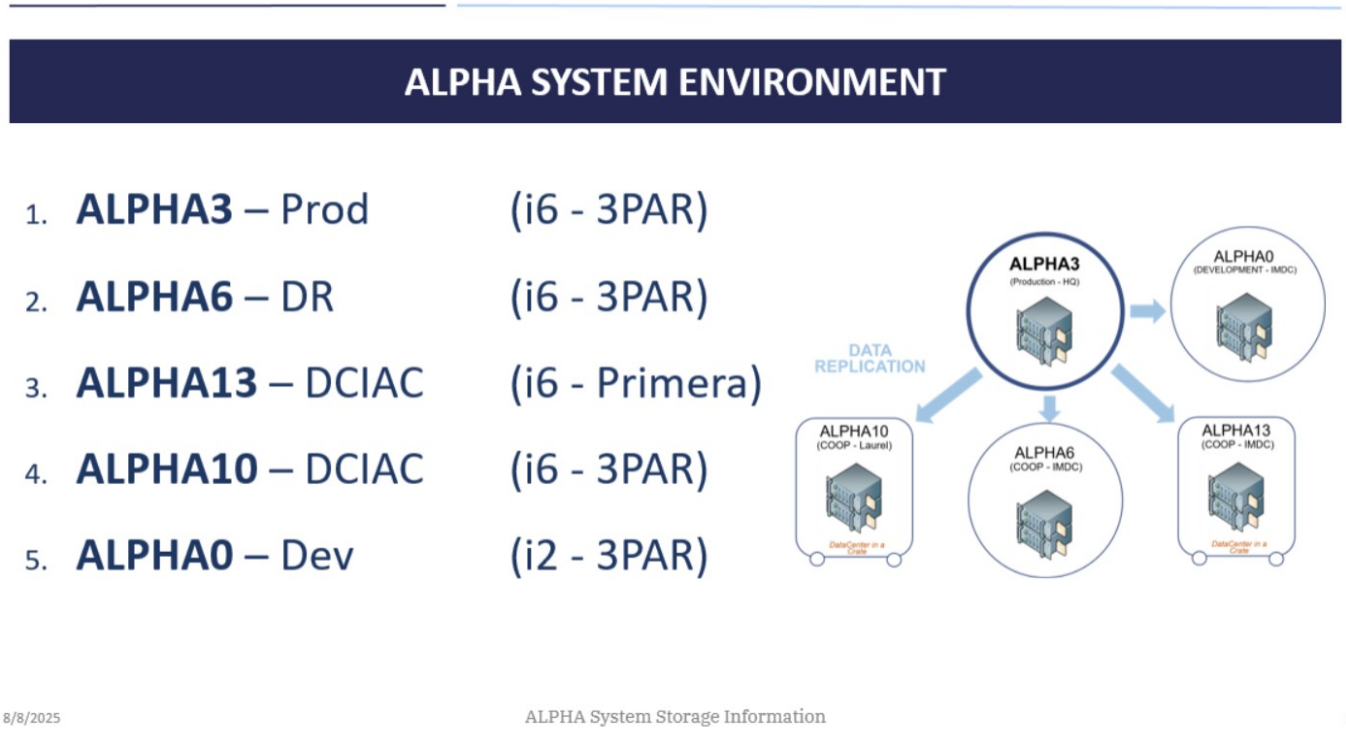


Figure 1 depicts the legacy Alpha estate as a set of distinct environments centered on ALPHA3 Production. It shows ALPHA6 as DR/backup, ALPHA10 and ALPHA13 as DCIAC recovery environments, ALPHA0 as development, and illustrates the existing production-data replication relationships that the GDPS replacement must preserve or improve.

2.3 Legacy Production Cluster Components

Figure 2

Legacy Production Cluster Components

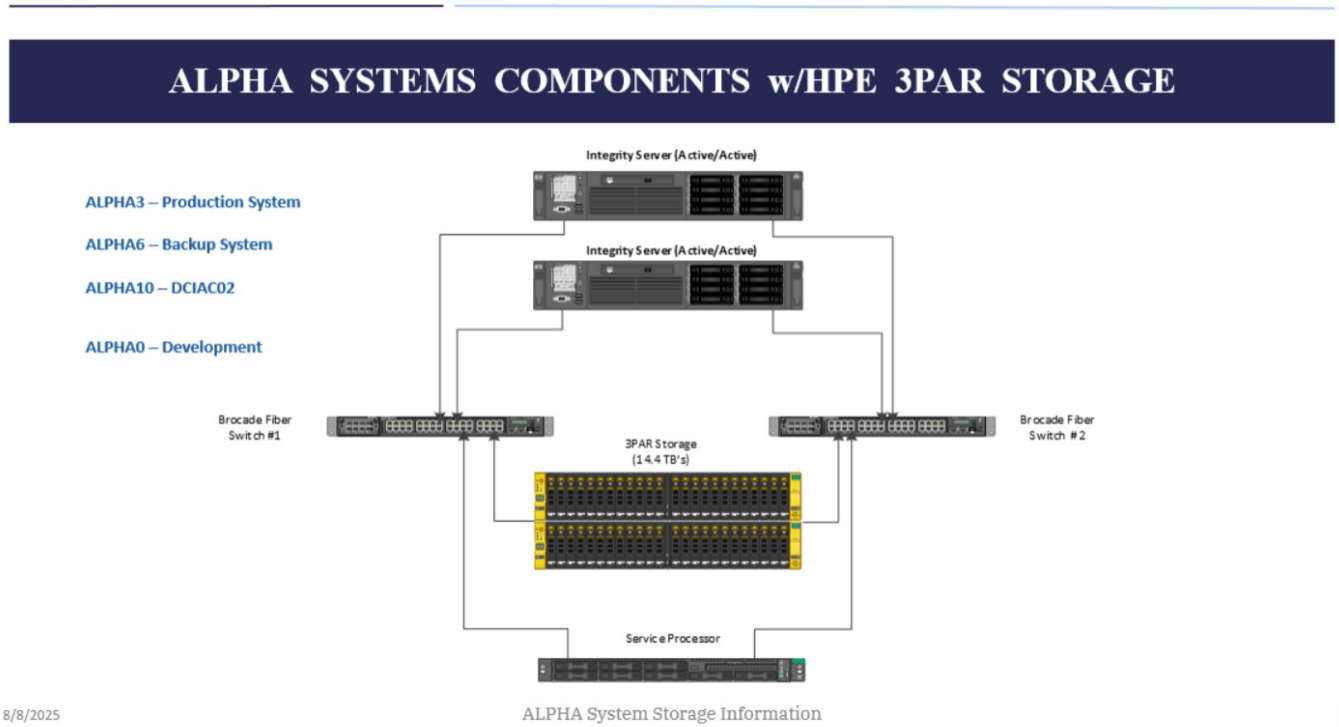


Figure 2 depicts the legacy production hardware pattern using two active/active Integrity servers connected through dual Brocade Fibre Channel switches to shared HPE 3PAR storage. The picture establishes the two-node clustered-compute, redundant-fabric, shared-storage pattern that serves as a direct baseline for the GDPS clustered file/service tier.

2.4 Legacy COOP Data Center in a Crate

Figure 3

Legacy COOP Data Center in a Crate

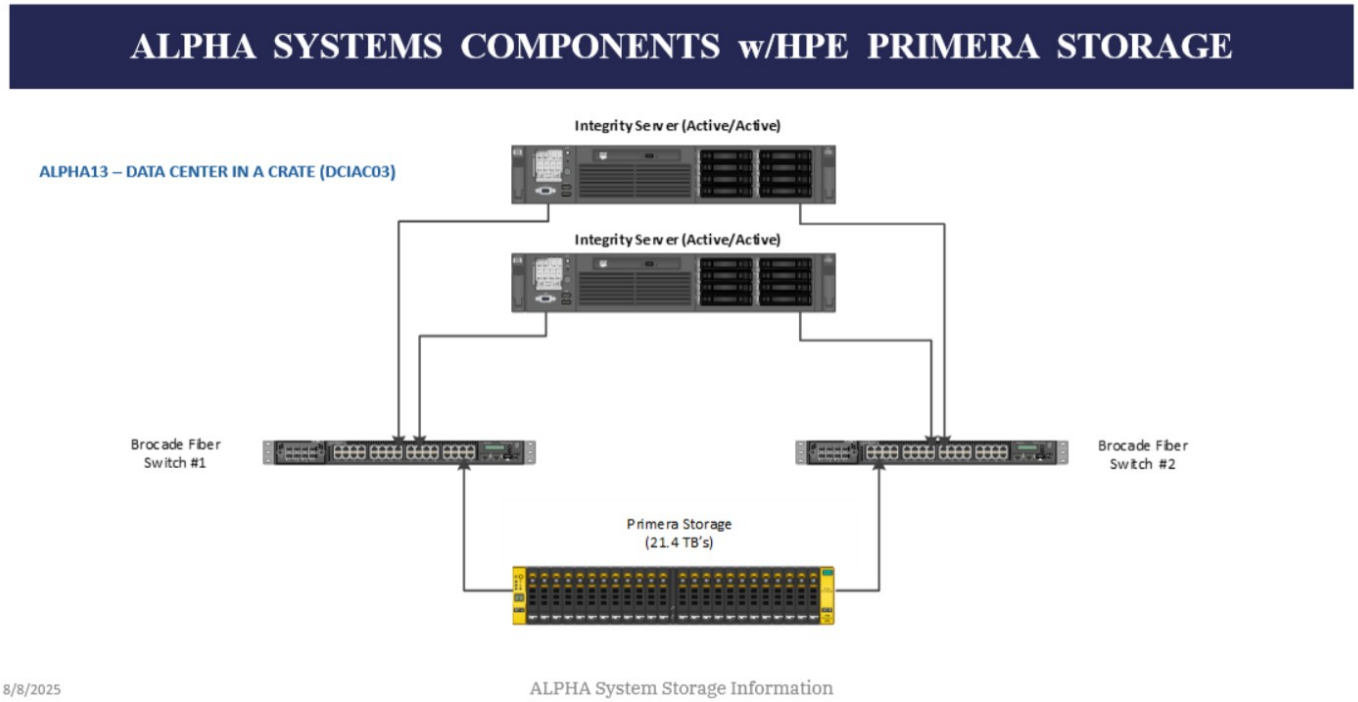


Figure 3 depicts the ALPHA13 Data Center in a Crate architecture. It shows two active/active Integrity servers, dual Brocade Fibre Channel switches, and shared HPE Primera storage, demonstrating that the COOP design is itself a two-node clustered recovery platform rather than a single recovery server.

2.5 Legacy Alpha3 Disk Volume Allocation

Figure 4

Legacy ALPHA3 Disk Volume Allocation

Disk Volume	ALPHA3	
	Used Space (GB)	**Total** Space (GB)
RAIDSET00	**150**	300
RAIDSET01	**400**	500
RAIDSET02	**485**	600
RAIDSET03	**50**	100
RAIDSET04	**20**	50
RAIDSET05	**20**	100
RAIDSET06	**60**	100
RAIDSET07	**5**	50
RAIDSET08	**10**	50

Figure 4 depicts the legacy ALPHA3 logical disk-volume allocation across RAIDSET00 through RAIDSET09. It shows the used and total capacity of each logical volume and provides the source values for the GDPS logical-volume mapping and migration-capacity traceability.

3. GDPS Replacement Hardware Architecture

Server and service identifiers use GDPS, the Alpha replacement system code; GIPS remains the department name. The physical file/service node names are architecture-controlled and are fixed in this document so implementation can be traced directly to the approved design. TEST uses HQVMGDPSFST1 and IMVMGDPSFST2; PROD uses HQVMGDPSFSP1 and IMVMGDPSFSP2; DR uses IMVMGDPSFSD1 and IMVMGDPSFSD2; COOP uses IMVMGDPSFSC1 and IMVMGDPSFSC2. TEST and PROD pairs are distributed across HQ and IM, DR is located at IMDC, and COOP remains in the designated DCIAC/CRATE recovery location. Detailed IP addressing is owned by the separate Network Design.

Figure 5

GDPS Replacement Environment Overview - DEV VM + Eight Physical Cluster Nodes

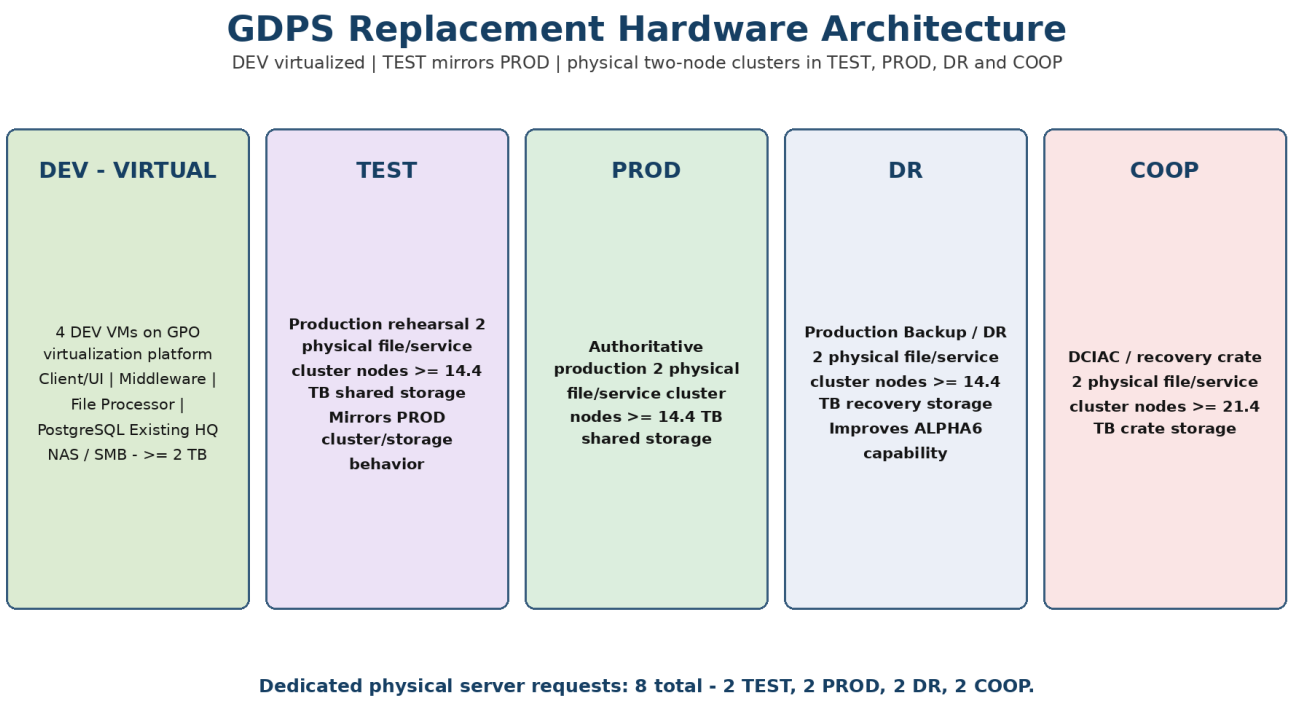


Figure 5 depicts the proposed GDPS environment structure across DEV, TEST, PROD, DR, and COOP. DEV is shown as a VM-based development stack on existing GPO virtualization infrastructure. TEST, PROD, DR, and COOP each use two dedicated physical file/service cluster nodes. TEST mirrors PROD, DR replaces and improves ALPHA6 Production Backup capability, and COOP preserves the DCIAC recovery model.

- DEV is a complete four-VM development stack using the existing GPO virtualization platform and external enterprise NAS storage.
- TEST is the new pre-production environment that Alpha does not have today. TEST mirrors the PROD hardware topology so failover, shared storage, replication, and deployment behavior can be validated before production.
- PROD replaces the ALPHA3 Production environment. DR is a separate two-node physical recovery environment that replaces and improves the ALPHA6 Production Backup System capability.
- COOP preserves the self-contained Data Center in a Crate recovery model represented by Alpha13/Alpha10.
- Global Admin is hosted on the same Client/UI IIS tier as the business client.

4. Hardware Server Matrices

4.1 DEV Environment

Table 7 defines the DEV virtual-machine baseline. It identifies the four DEV VMs, their application roles, vCPU and memory targets, local virtual-disk allocations, external/shared-storage dependencies, and single-instance development availability. DEV does not require a dedicated physical server request under this baseline.

Table 7

DEV Environment Virtual Machine Matrix

Server Name	Role	vCPU	RAM	Local Storage	External / Shared Storage	Availability
HQVMGDPSCLID1	Client/UI + Global Admin + IIS	4	16 GB	120 GB OS + 100 GB app/log	N/A	DEV VM - single instance
HQVMGDPSMWDD1	Middleware API + Issuer + Delivery Worker	8	32 GB	120 GB OS + 200 GB app/log	\ \hqnas\PlantOps\GDPS\ DEV\...	DEV VM - single instance
HQVMGDPSFPDD1	File Processor	8	32 GB	120 GB OS + 250 GB scratch/temp	\ \hqnas\PlantOps\GDPS\ DEV\...	DEV VM - single instance
HQVMGDPSPGDD1	PostgreSQL	8	32 GB	120 GB OS	1 TB enterprise DB storage	DEV VM - single instance
DEV SHARED STORAGE	External GDPS file storage				2 TB minimum allocation on existing HQ NAS	External enterprise NAS

Figure 6

GDPS DEV Virtual Machine Layout

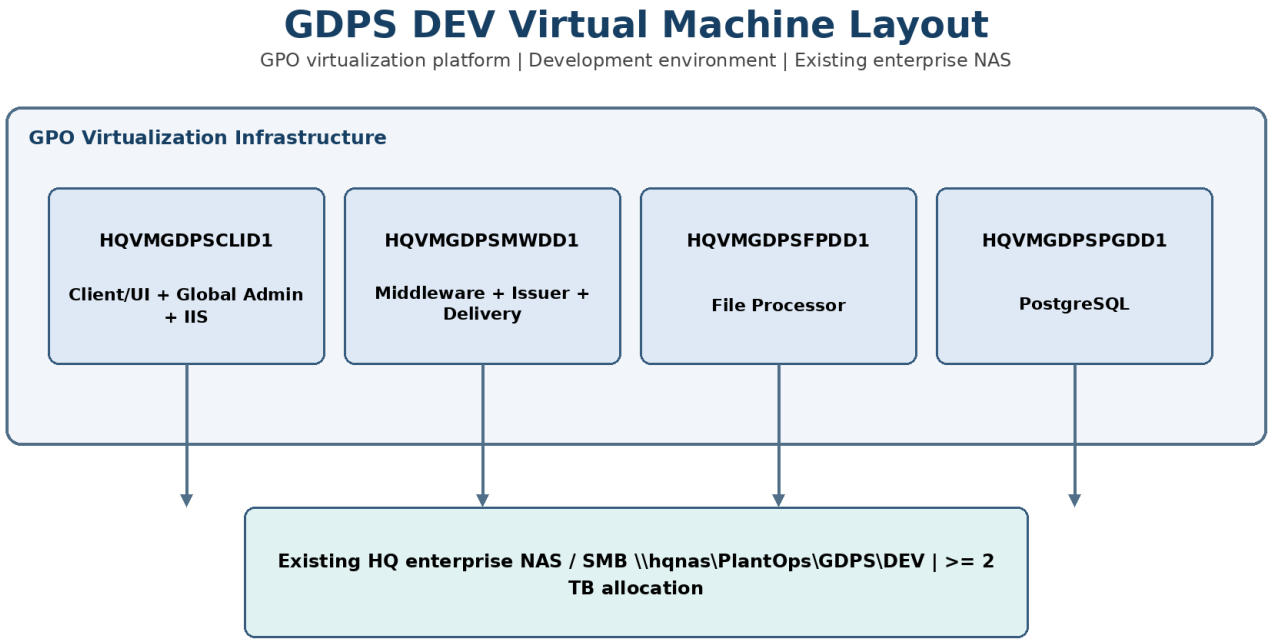


Figure 6 depicts the GDPS DEV virtual-machine layout. It shows four separate DEV VMs for Client/UI, Middleware, File Processor, and PostgreSQL running on the existing GPO virtualization platform and using external enterprise NAS storage for shared file data. DEV virtualization is an intentional efficiency improvement and does not change the physical two-node baseline used for TEST, PROD, DR, and COOP.

4.2 TEST Environment (Production Mirror)

Table 8 defines the TEST environment as a production mirror. It shows the logical application/database roles and the two dedicated physical file/service cluster nodes split between the HQ and IM data centers. The two physical TEST nodes use a minimum 128 GB RAM baseline and production-equivalent shared storage so node failover, full data-center failover, storage, replication, and cutover behavior can be rehearsed before PROD.

Table 8

TEST Environment Hardware Server Matrix

Server Name	Role	vCPU	RAM	Local Storage	External / Shared Storage	Availability
HQVMGDPSCLIT1	Client/UI + Global Admin + IIS — Primary	4	16 GB	120 GB OS + 100 GB app/log	N/A	Failover pair
IMVMGDPSCLIT2	Client/UI + Global Admin + IIS — Failover	4	16 GB	120 GB OS + 100 GB app/log	N/A	Failover pair
HQVMGDPSMWDT1	Middleware API + Issuer + Delivery Worker —	8	32 GB	120 GB OS + 200 GB app/log	TEST shared storage	Failover pair

	Primary					
IMVMGDPSMWDT2	Middleware API + Issuer + Delivery Worker — Failover	8	32 GB	120 GB OS + 200 GB app/log	TEST shared storage	Failover pair
HQVMGDPSFPDT1	File Processor — Primary	8	32 GB	120 GB OS + 250 GB scratch/temp	TEST shared storage	Failover pair
IMVMGDPSFPDT2	File Processor — Failover	8	32 GB	120 GB OS + 250 GB scratch/temp	TEST shared storage	Failover pair
HQVMGDPSPGDT1	PostgreSQL — Primary	8	64 GB	120 GB OS	2 TB enterprise DB storage	HA pair
IMVMGDPSPGDT2	PostgreSQL — Failover	8	64 GB	120 GB OS	2 TB enterprise DB storage	HA pair
HQVMGDPSFST1	File Cluster Node 1	Physical / CDW-G engineered	128 GB minimum	Standard Windows Server OS drive; no authoritative business data	Shared enterprise storage	2-node physical WSFC
IMVMGDPSFST2	File Cluster Node 2	Physical / CDW-G engineered	128 GB minimum	Standard Windows Server OS drive; no authoritative business data	Shared enterprise storage	2-node physical WSFC
TEST STORAGE	Clustered GDPS shared file storage				14.4 TB usable minimum; redundant fabric/multipath	Mirrors PROD

Figure 7

GDPS TEST Hardware Layout

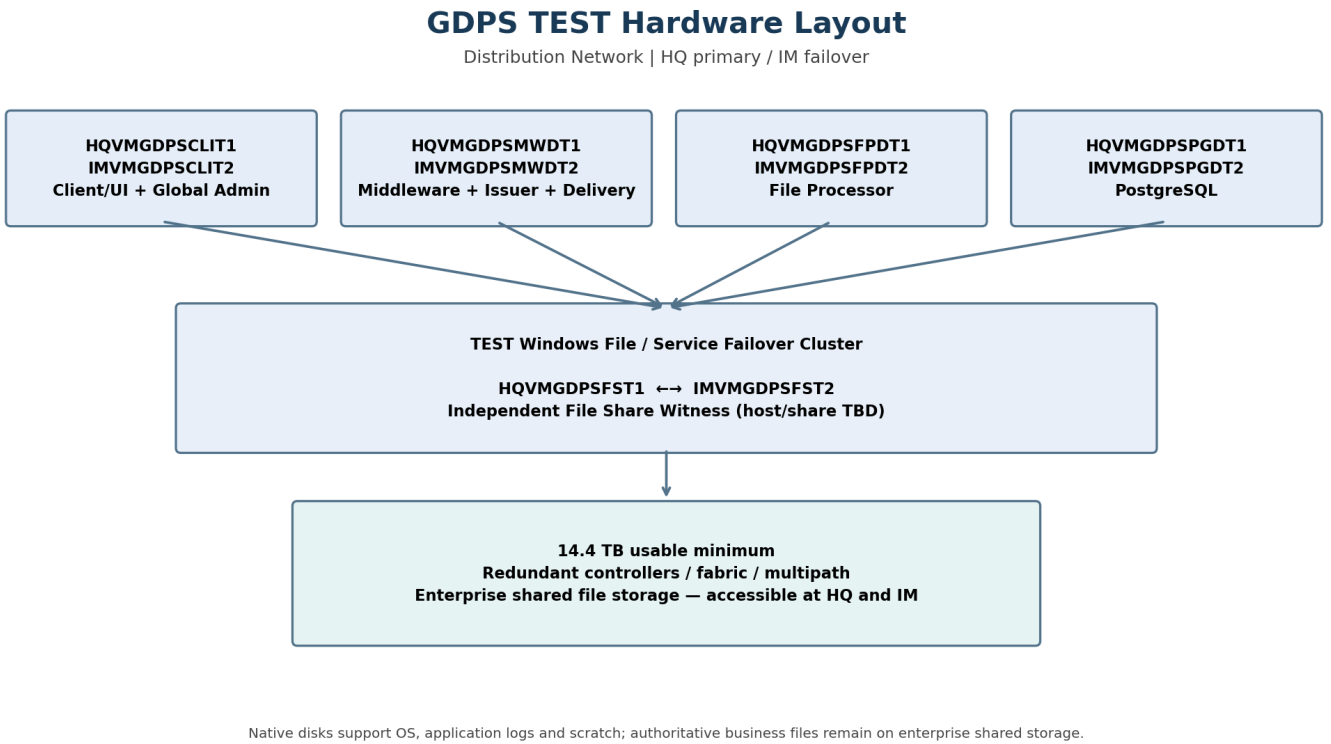


Figure 7 depicts the GDPS TEST topology and emphasizes that TEST mirrors PROD. The physical file/service pair is intentionally split across the separate HQ and IM data centers. This allows TEST to exercise both ordinary node failure and complete site-loss scenarios, using production-equivalent shared storage, quorum, replication, deployment, and recovery behavior before production changes are made.

4.3 PROD, DR, and COOP Environments

Table 9 presents the server and storage requirements for PROD, DR, and COOP. It identifies the logical application/database roles already defined for PROD and COOP and, most importantly for this hardware acquisition, the dedicated two-node physical file/service clusters. DR is added as a separate two-node physical recovery environment aligned to the confirmed ALPHA6 Production Backup baseline.

Table 9

PROD, DR, and COOP Environment Hardware Server Matrix

Server Name	Role	vCPU	RAM	Local Storage	External / Shared Storage	Availability
PROD — HQVMGDPSCLIP1	Client/UI + Global Admin + IIS — Primary	4	16 GB	120 GB OS + 100 GB app/log	N/A	Production failover
PROD —	Client/UI + Global Admin	4	16 GB	120 GB OS + 100 GB	N/A	Production failover

IMVMGDPSCLIP2	+ IIS — Failover			app/log		
PROD — HQVMGDPSMWDP1	Middleware API + Issuer + Delivery Worker — Primary	8	32 GB	120 GB OS + 200 GB app/log	PROD shared storage	Production failover
PROD — IMVMGDPSMWDP2	Middleware API + Issuer + Delivery Worker — Failover	8	32 GB	120 GB OS + 200 GB app/log	PROD shared storage	Production failover
PROD — HQVMGDPSFPDP1	File Processor — Primary	8	32 GB	120 GB OS + 250 GB scratch/temp	PROD shared storage	Production failover
PROD — IMVMGDPSFPDP2	File Processor — Failover	8	32 GB	120 GB OS + 250 GB scratch/temp	PROD shared storage	Production failover
PROD — HQVMGDPSPGDP1	PostgreSQL — Primary	8	64 GB	120 GB OS	2 TB enterprise DB storage	Production HA
PROD — IMVMGDPSPGDP2	PostgreSQL — Failover	8	64 GB	120 GB OS	2 TB enterprise DB storage	Production HA
HQVMGDPSFSP1	File Cluster Node 1	Physical / CDW-G engineered	128 GB minimum	Standard Windows Server OS drive; no authoritative business data	Shared enterprise storage	2-node physical WSFC
IMVMGDPSFSP2	File Cluster Node 2	Physical / CDW-G engineered	128 GB minimum	Standard Windows Server OS drive; no authoritative business data	Shared enterprise storage	2-node physical WSFC
PROD STORAGE	Clustered GDPS shared file storage				14.4 TB usable minimum; redundant fabric/multipath	Replicates to COOP
COOP — IMVMGDPSCLIC1	Client/UI + Global Admin + IIS	4	16 GB	120 GB OS + 100 GB app/log	N/A	Crate recovery stack
COOP — IMVMGDPSMWDC1	Middleware API + Issuer + Delivery Worker	8	32 GB	120 GB OS + 200 GB app/log	COOP crate storage	Crate recovery stack
COOP — IMVMGDPSFPDC1	File Processor	8	32 GB	120 GB OS + 250 GB scratch/temp	COOP crate storage	Crate recovery stack
COOP — IMVMGDPSPGDC1	PostgreSQL	8	64 GB	120 GB OS	2 TB enterprise DB storage	Single recovery DB
IMVMGDPSFSC1	File Cluster Node 1	Physical / CDW-G engineered	128 GB minimum	Standard Windows Server OS drive; no authoritative business data	Shared crate storage	2-node physical WSFC
IMVMGDPSFSC2	File Cluster Node 2	Physical / CDW-G engineered	128 GB minimum	Standard Windows Server OS drive; no authoritative business data	Shared crate storage	2-node physical WSFC
COOP STORAGE	Crate GDPS shared file storage				21.4 TB usable minimum; redundant fabric/multipath	Receives PROD replication
IMVMGDPSFSD1	File/Service Cluster Node 1	Physical / CDW-G engineered	128 GB minimum	Standard Windows Server OS drive; no authoritative business data	DR shared enterprise storage	2-node physical WSFC
IMVMGDPSFSD2	File/Service Cluster Node 2	Physical / CDW-G engineered	128 GB minimum	Standard Windows Server OS drive; no authoritative business data	DR shared enterprise storage	2-node physical WSFC
DR STORAGE	GDPS disaster-recovery shared file storage				14.4 TB usable minimum; redundant	Receives PROD recovery replication

					fabric/multipath	

Figure 8

GDPS PROD Hardware Layout

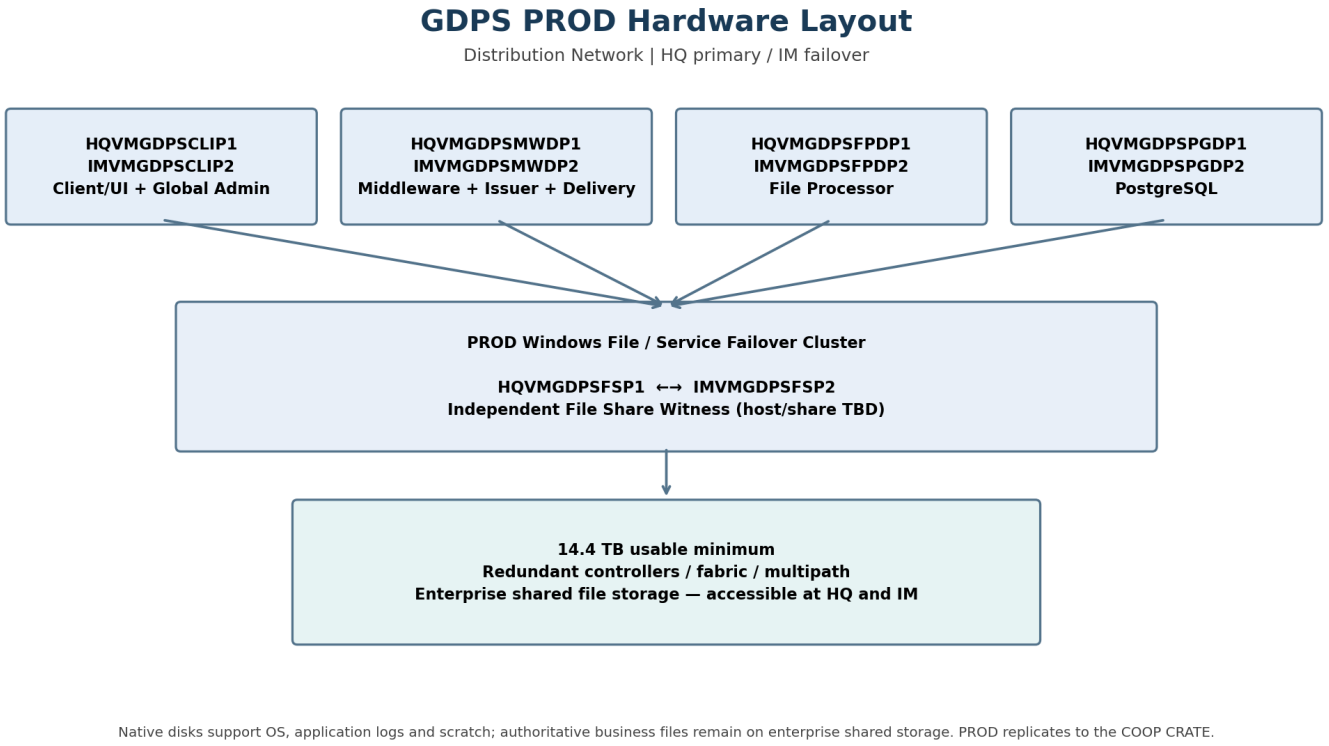


Figure 8 depicts the GDPS PROD topology. It shows the two physical Windows file/service nodes split across the separate HQ and IM data centers, together with shared enterprise storage and the dependent application/database services. This topology replaces the ALPHA3 production capability while improving resilience so service can fail over to the alternate data center, not merely to another server in the same site, and while supporting replication to recovery infrastructure.

Figure 9

GDPS COOP Crate Hardware Layout

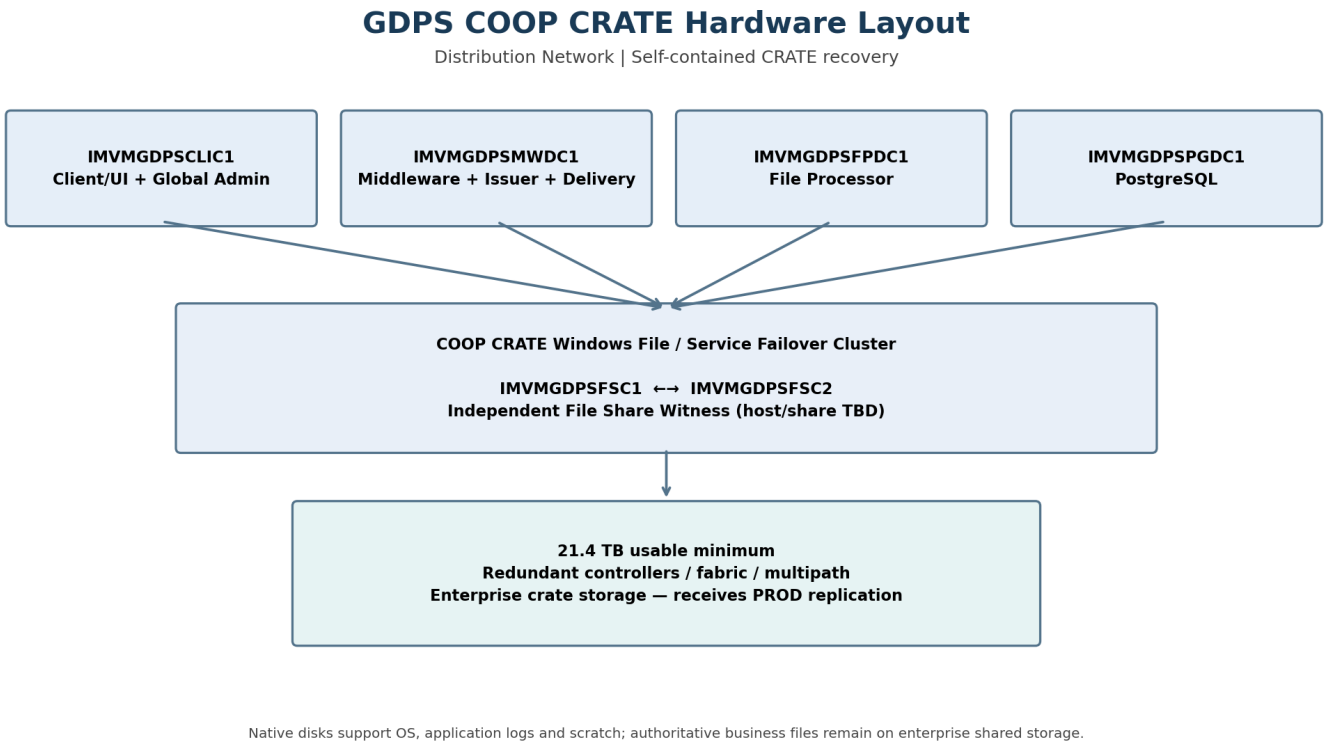


Figure 9 depicts the GDPS COOP crate topology. It shows the recovery application stack, the two-node Windows file cluster, and independent crate storage needed for continued operation when the primary production site is unavailable.

5. Disk Volume / GDPS Volume Mapping

The following table preserves the existing Alpha RAIDSET00–RAIDSET09 logical-volume allocation as the capacity and traceability baseline. The GDPS logical volume identifiers provide a direct planning equivalent for the replacement shared-storage design. Final Windows/NAS mount or share paths are defined during storage configuration.

Table 10 maps the legacy Alpha RAIDSET00 through RAIDSET09 volumes to planned GDPS logical volumes. It preserves the known used, provisioned, and free-capacity values so the replacement can maintain traceability to the current OpenVMS disk layout without treating the logical allocation as the full physical-array capacity.

Table 10

Legacy Alpha to GDPS Disk Volume Mapping

Legacy Alpha Volume	GDPS Logical Volume	Used	Provisioned	Free
RAIDSET00	GDPS_VOL_00	150 GB	300 GB	150 GB

RAIDSET01	GDPS_VOL_01	400 GB	500 GB	100 GB
RAIDSET02	GDPS_VOL_02	485 GB	600 GB	115 GB
RAIDSET03	GDPS_VOL_03	50 GB	100 GB	50 GB
RAIDSET04	GDPS_VOL_04	20 GB	50 GB	30 GB
RAIDSET05	GDPS_VOL_05	20 GB	100 GB	80 GB
RAIDSET06	GDPS_VOL_06	60 GB	100 GB	40 GB
RAIDSET07	GDPS_VOL_07	5 GB	50 GB	45 GB
RAIDSET08	GDPS_VOL_08	10 GB	50 GB	40 GB
RAIDSET09	GDPS_VOL_09	5 GB	100 GB	95 GB
TOTAL	—	1,205 GB	1,950 GB	745 GB

6. Shared Storage Baseline

Table 11 summarizes the shared-storage baseline by environment. DEV continues to use existing enterprise NAS. TEST and PROD retain matching 14.4 TB minimum shared-storage baselines, DR adds a production-capable 14.4 TB minimum recovery target, and COOP retains the larger 21.4 TB crate baseline. Because TEST and PROD span the separate HQ and IM data centers, their storage architecture must remain available or recoverable when either site is lost; CDW-G/GPO Storage must engineer the appropriate cross-site presentation or replication mechanism. TEST, PROD, DR, and COOP require redundant controllers/fabric/multipath as applicable.

Table 11

Shared Storage Baseline

Environment	Storage Source / Type	Minimum Usable Capacity	Availability Characteristics	Purpose
DEV	Existing HQ NAS / SMB	2 TB minimum allocated	External enterprise NAS	\\hqnas\PlantOps\GDPS\DEV\...
TEST	Production-equivalent enterprise shared storage	14.4 TB usable minimum	Redundant controllers/fabric/multipath	Same storage architecture as PROD
PROD	Enterprise shared storage replacing ALPHA3 production storage capability	14.4 TB usable minimum	Redundant controllers/fabric/multipath	Authoritative production storage; replication source to DR and COOP
COOP	Enterprise shared storage within recovery crate	21.4 TB usable minimum	Redundant controllers/fabric/multipath	Receives production replication
DR	Production-recovery enterprise shared storage	14.4 TB usable minimum	Redundant controllers/fabric/multipath	Receives PROD recovery replication; supports ALPHA6 replacement capability

7. Replication and Failover Architecture

Figure 10

GDPS Data Replication, DR, and COOP Recovery Topology

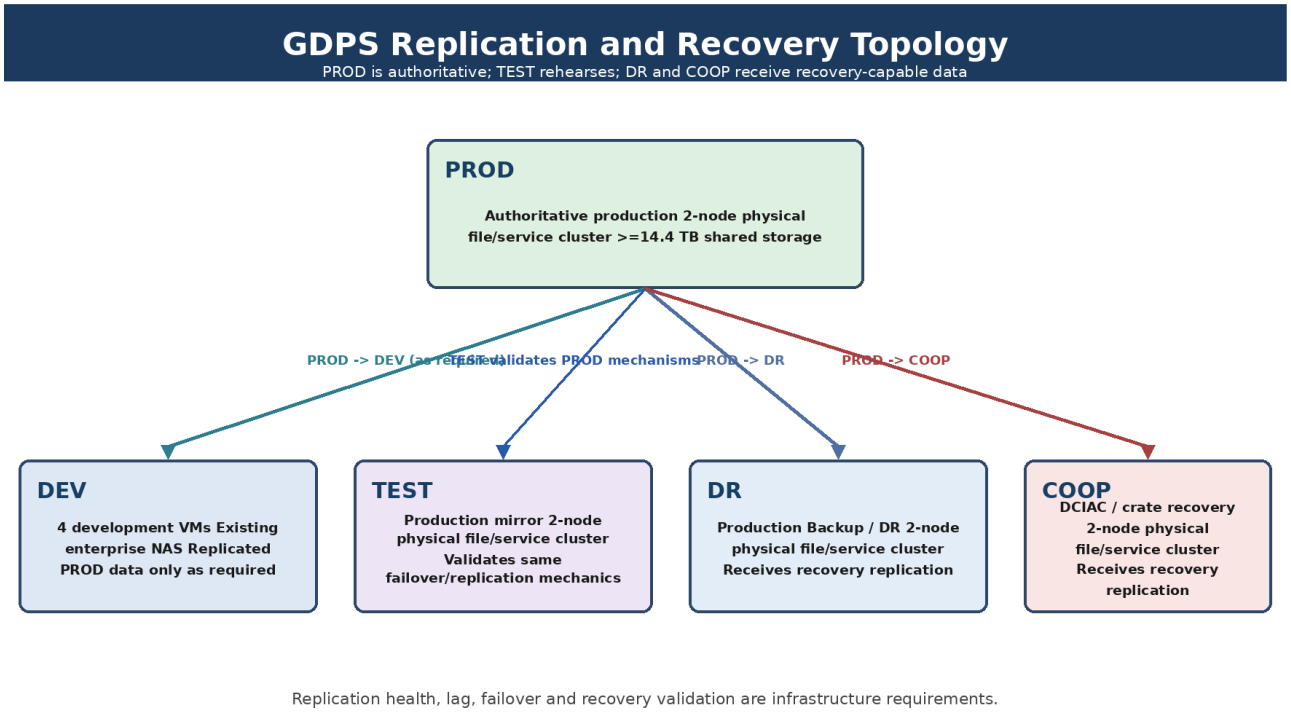


Figure 10 depicts the GDPS data-replication and recovery relationships. PROD remains authoritative. TEST validates the same failover and replication mechanisms without using production identities. DR receives production recovery data to replace and improve the ALPHA6 Production Backup capability, while COOP receives production data for DCIAC recovery. DEV receives replicated production data only where the current development model requires it.

- PROD remains the authoritative production data source.
- PROD data replicates to DR and COOP so either approved recovery capability can be prepared for disruption scenarios.
- DEV continues to receive replicated production data where the current Alpha development model requires that data.
- TEST uses the same clustered storage and replication mechanisms as PROD so production failover and recovery can be validated before deployment; TEST never owns live production identities.
- Replication status, lag, failure alerting, failover and recovery validation are infrastructure requirements.

8. Legacy-to-GDPS Hardware Mapping

Table 12 provides the direct current-to-future hardware mapping. It shows which GDPS environment replaces each legacy Alpha capability, the replacement hardware pattern, and whether the result preserves an existing function or adds an improvement such as the new production-mirroring TEST environment.

Table 12

Legacy-to-GDPS Hardware Mapping

Legacy Capability	GDPS Replacement	Replacement Hardware Pattern	Preserved / Added Capability
Alpha0 / DCIAC1	GDPS DEV	Four-VM development stack + external enterprise NAS	Preserve development functionality while removing unnecessary dedicated physical DEV hardware
Alpha3	GDPS PROD	Two-node application tiers + two-node file cluster + shared enterprise storage	Preserve primary production clustered-service behavior
Alpha6	GDPS DR	Two-node physical file/service cluster + production-recovery shared storage	Preserve and improve the current ALPHA6 Production Backup / DR capability
Alpha10 / DCIAC2	GDPS COOP / DCIAC recovery capability	Two-node recovery/file-service capability	Preserve the confirmed DCIAC02 two-node recovery baseline
Alpha13 / DCIAC3	GDPS COOP crate	Self-contained application stack + two-node file cluster + crate storage	Preserve the confirmed DCIAC03 two-node Data Center In-a-Crate capability and replicated production data
No dedicated Alpha TEST tier	GDPS TEST	Full PROD mirror	New pre-production environment for validating PROD hardware, clustering, failover and replication

9. Provisioning Summary

Table 13 consolidates the logical provisioning quantities across DEV, TEST, PROD, DR, and COOP. DEV uses four VMs. The dedicated physical-server acquisition described in this hardware baseline is the file/service cluster tier: two physical nodes each for TEST, PROD, DR, and COOP, for a total of eight physical servers.

Table 13

GDPS Provisioning Summary

Component	DEV	TEST	PROD	DR	COOP
Client/UI + Global Admin	1 VM	2 logical roles	2 logical roles	Per DR application design	1 logical role
Middleware + Issuer + Delivery	1 VM	2 logical roles	2 logical roles	Per DR application design	1 logical role
File Processor	1 VM	2 logical roles	2 logical roles	Per DR application design	1 logical role
PostgreSQL	1 VM	2 logical roles	2 logical roles	Per DR application design	1 logical role
Physical File/Service Cluster Nodes	0	2	2	2	2
Shared File Storage	2 TB NAS allocation	14.4 TB min	14.4 TB min	14.4 TB min	21.4 TB min
Replication	As current DEV model requires	Mirrors PROD mechanisms	Authoritative source	Receives PROD recovery replica	Receives PROD recovery replica

9.1 Internal GPO Physical Server Request Baseline

Table 17 identifies the eight dedicated physical servers that require individual GPO IT Service Hub New Server requests. DEV does not appear because its four development roles are VMs. Each physical request is one member of a matched two-node WSFC file/service pair and must be provisioned in coordination with its partner node and environment storage.

Table 17

Eight Physical Server Requests for GDPS Clustered File/Service Tier

Naming control. The server names in Table 17 are part of the approved GDPS hardware architecture, not placeholders for Infrastructure to rename. Each GPO New Server request and each CDW-G/GPO implementation artifact shall use the same identifier. A proposed rename is an architecture change and must be reconciled in this master document before provisioning.

Request	Environment	Physical Server Name	Placement	OS	RAM	Network	Shared Storage
SR-01	TEST	HQVMGDPSFST1	HQ Data Center	Windows Server 2025	128 GB minimum	Server Distribution	TEST >=14.4 TB
SR-02	TEST	IMVMGDPSFST2	IM Data Center	Windows Server 2025	128 GB minimum	Server Distribution	TEST >=14.4 TB
SR-03	PROD	HQVMGDPSFSP1	HQ Data Center	Windows Server 2025	128 GB minimum	Server Distribution	PROD >=14.4 TB
SR-04	PROD	IMVMGDPSFSP2	IM Data Center	Windows Server 2025	128 GB minimum	Server Distribution	PROD >=14.4 TB
SR-05	DR	IMVMGDPSFSD1	IMDC	Windows Server 2025	128 GB minimum	Server Distribution	DR >=14.4 TB
SR-06	DR	IMVMGDPSFSD2	IMDC	Windows Server 2025	128 GB minimum	Server Distribution	DR >=14.4 TB
SR-07	COOP	IMVMGDPSFSC1	DCIAC / CRATE	Windows Server 2025	128 GB minimum	Server Distribution	COOP >=21.4 TB
SR-08	COOP	IMVMGDPSFSC2	DCIAC / CRATE	Windows Server 2025	128 GB minimum	Server Distribution	COOP >=21.4 TB

The physical-host names are fixed in Table 17 and are part of the GDPS architecture baseline. Final IP addresses, DNS records, VLAN assignments, security boundaries, and the exact COOP ServiceNow data-center catalog selection are assigned by the appropriate GPO Network, Security, and Infrastructure teams. Those teams shall implement the documented server names rather than selecting alternate hostnames.

10. Infrastructure Guardrails

- Do not place authoritative GDPS business file storage on application-server local disks.
- Do not reduce TEST, PROD, DR, or COOP clustered file/service capability to a single non-clustered file server.
- Do not remove existing replication behavior during the technology port.
- TEST must continue to reflect PROD hardware topology, storage design, cross-data-center placement, site-level failover behavior, and replication mechanisms.
- The COOP crate must remain independently operable if the primary production environment is unavailable.
- Hardware implementation changes must not introduce new POLLVAX business rules.

- Do not reduce a confirmed legacy two-node cluster capability to a single replacement file/service node; any topology change must maintain or improve the existing availability outcome.

11. GDPS Windows File Cluster Standard

TEST, PROD, DR, and COOP file-service tiers shall use Windows Server Failover Clustering (WSFC) with two dedicated physical file/service nodes per environment. TEST and PROD are cross-data-center clusters with one node at HQ and one node at IM, allowing the service to move to the alternate data center if an entire site is unavailable. This eight-server physical baseline preserves the proven two-node Alpha availability pattern while DEV is intentionally hosted as virtual machines.

- Cluster type: two-node physical Windows Server Failover Cluster for TEST, PROD, DR, and COOP file/service tiers.
- File-server model: clustered general-purpose SMB file service used to preserve the existing Alpha file-share interfaces.
- Quorum: File Share Witness.
- Witness placement: the witness share must be hosted on infrastructure independent of the clustered GDPS data storage and must not create a dependency on only HQ or only IM. For the cross-data-center TEST and PROD clusters, the witness must remain reachable when either data center is unavailable so quorum can be established at the surviving site. The exact witness host/path remains an infrastructure design decision, but the site-failure requirement is mandatory.
- Storage: shared enterprise storage presented to both cluster nodes with redundant storage paths/fabric and MPIO.
- The physical GDPS file/service node hostnames are fixed architecture identifiers and shall be provisioned exactly as documented: HQVMGDPSFST1 (TEST at HQ), IMVMGDPSFST2 (TEST at IM), HQVMGDPSFSP1 (PROD at HQ), IMVMGDPSFSP2 (PROD at IM), IMVMGDPSFSD1 and IMVMGDPSFSD2 (DR at IMDC), and IMVMGDPSFSC1 and IMVMGDPSFSC2 (COOP/DCIAC). The HQ/IM prefix identifies the data-center placement and is therefore an availability attribute, not cosmetic naming. Legacy Alpha names are not used as physical Windows hostnames.
- The cluster administrative identity is separate from the SMB client-access identities consumed by legacy applications.

11.1 File Share Witness Baseline

Table 14 defines the quorum baseline for the two-node TEST, PROD, DR, and COOP physical Windows file/service clusters. Each cluster uses an independent File Share Witness so the two-node design has a third quorum vote without placing the witness on clustered business-data storage.

Table 14

File Share Witness Baseline

Environment	Cluster Nodes	Quorum Model	Witness Location	Placement Requirement
TEST	HQVMGDPSFST1 / IMVMGDPSFST2	File Share Witness	Witness host/share TBD	Independent of TEST clustered data storage
PROD	HQVMGDPSFSP1 / IMVMGDPSFSP2	File Share Witness	Witness host/share TBD	Independent of PROD clustered data storage
COOP	IMVMGDPSFSC1 / IMVMGDPSFSC2	File Share Witness	Witness host/share TBD	Must remain reachable during COOP operation and be independent of crate clustered data storage
DR	IMVMGDPSFSD1 / IMVMGDPSFSD2	File Share Witness	Witness host/share TBD	Independent of DR clustered data storage; reachable during DR operation

12. Legacy SMB Namespace Preservation

Legacy Windows applications currently consume Alpha-hosted files through UNC paths whose server portion is the Alpha system name, for example \\alpha0\<share>\<subfolder>, \\alpha3\<share>\<subfolder>, and \\alpha6\<share>\<subfolder>. Those applications cannot be changed during the GDPS hardware migration. The GDPS production file cluster must therefore preserve these UNC interfaces.

The Alpha names shall be treated as SMB client-access service identities rather than as the physical names of GDPS cluster nodes. This allows the GDPS nodes to retain stable infrastructure hostnames while legacy applications continue to use the original UNC paths.

Table 15 identifies the SMB identities used during migration and steady-state operations. It distinguishes permanent GDPS physical node names from temporary migration names and final legacy-compatible client-access identities, making clear when each name is permitted and what operational purpose it serves.

Table 15

Legacy SMB Identity Baseline

Identity Type	Name	When Used	Purpose
Physical PROD node	HQVMGDPSFSP1	Permanent	WSFC physical file/service cluster node 1
Physical PROD node	IMVMGDPSFSP2	Permanent	WSFC physical file/service cluster node 2
Temporary migration SMB identity	alpha0-1	Pre-cutover	Staging/validation while legacy alpha0 remains active
Temporary migration SMB identity	alpha3-1	Pre-cutover	Staging/validation while legacy alpha3 remains active
Temporary migration SMB identity	alpha6-1	Pre-cutover	Staging/validation while legacy alpha6 remains active
Legacy production SMB identity	alpha0	Post-cutover	Presented by GDPS file cluster; preserves legacy UNC paths
Legacy production SMB identity	alpha3	Post-cutover	Presented by GDPS file cluster; preserves legacy UNC paths
Legacy production SMB identity	alpha6	Post-cutover	Presented by GDPS file cluster; preserves legacy UNC paths

12.1 SMB Namespace Cutover Sequence

1. Build and validate the GDPS production file cluster using the permanent GDPS physical node names.
2. Stage and validate migrated shares through temporary SMB identities such as alpha0-1, alpha3-1, and alpha6-1 while the existing Alpha systems remain active.
3. Freeze legacy writes/processing at the production cutover point.
4. Complete the final data replication/synchronization pass and verify the synchronized dataset.
5. Disable the legacy Alpha SMB/network identities before the same production identities are activated on GDPS.
6. Activate alpha0, alpha3, and alpha6 as clustered SMB client-access identities on the GDPS production file cluster.
7. Validate DNS/name registration, SMB access, share names, permissions, and authentication from representative unchanged legacy applications.
8. Release GDPS production file services to the legacy consumers.

12.2 SMB Usage Boundary

Table 16 defines where SMB is required and where it is not. It preserves SMB for legacy file interfaces, clustered file services, configured File Processor paths, and delivery compatibility while preventing SMB from becoming the general internal communication mechanism for Middleware, Issuer, PostgreSQL, or browser-based application traffic.

Table 16

SMB Usage Boundary

GDPS / Legacy Area	SMB Requirement	Reason
Legacy application access to \\alpha0, \\alpha3, \\alpha6	SMB required	Preserved external interface
GDPS clustered file service	SMB required	Provides the preserved UNC namespaces
File Processor access to configured shared folders	SMB/UNC where configured	Uses external/shared file storage; does not make local disk authoritative
Delivery to preserved legacy shares	SMB required where destination is a legacy file share	Delivery interface requirement
Middleware API	SMB not required	HTTP/API runtime
Authentication Issuer	SMB not required	Authentication/token runtime
PostgreSQL	SMB not required	Database runtime
Client/UI	SMB not required for normal application operation	Web/IIS tier

Figure 11

GDPS PROD File Cluster and Legacy SMB Namespace

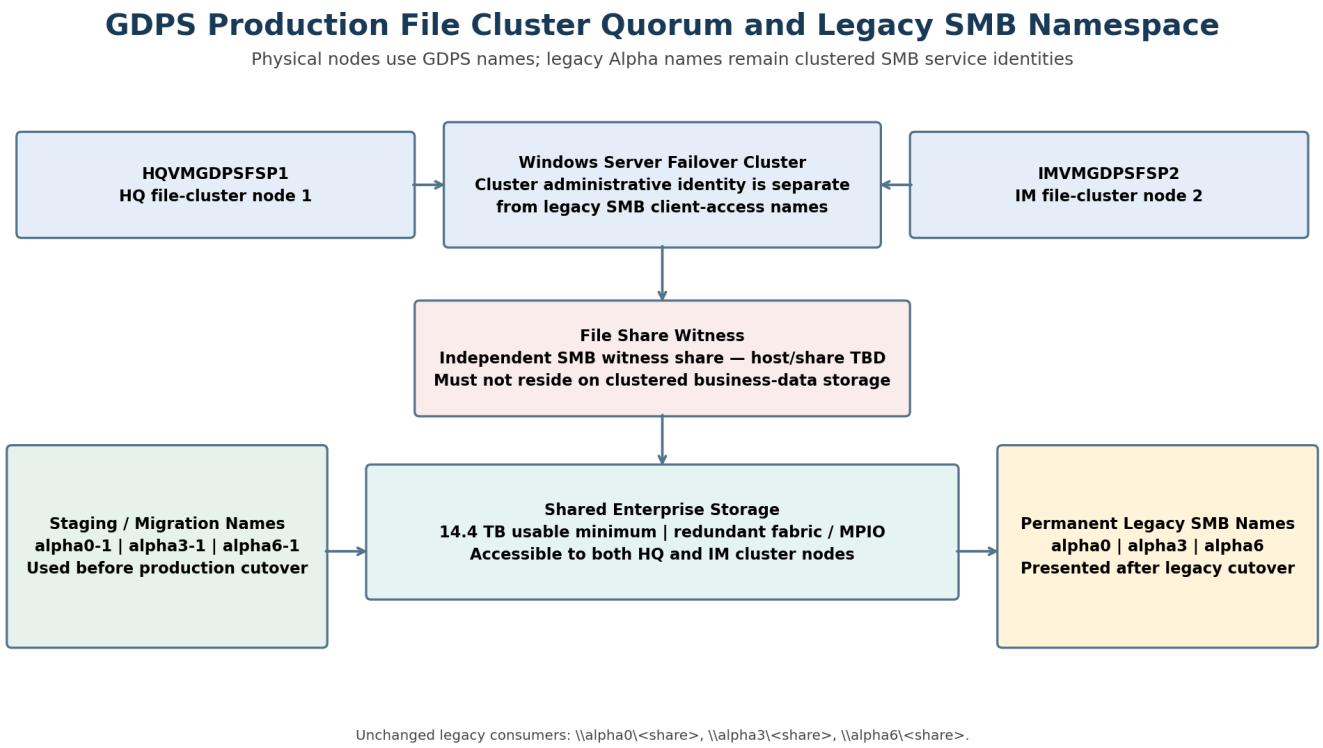


Figure 11 depicts the production file-cluster namespace and quorum design. It shows the permanent GDPS cluster nodes, the independent File Share Witness, temporary migration identities used before cutover, and the legacy-compatible alpha0, alpha3, and alpha6 SMB client-access identities that are activated only after the old Alpha identities are released.

13. Additional Infrastructure Guardrails

- Do not rename the physical GDPS file-cluster nodes to alpha0, alpha3, or alpha6.
- Do not have the legacy Alpha environment and the GDPS production cluster simultaneously advertise the same production SMB client-access identity.
- Temporary '-1' SMB identities are staging/migration identities only and are not the final production interface.
- The File Share Witness is quorum infrastructure only; it does not host GDPS business data.
- The File Share Witness must remain independent of the shared enterprise storage used by the file cluster.
- SMB is retained where required for legacy interoperability and clustered shared-file access; it is not the internal communication mechanism for all GDPS runtimes.